Kaleb Bruwer
kaleb.bruwer@gmail.com
(+27)71 441 8189
Github
website

Kaleb Bruwer

Software Engineer

Summary Software Engineer specializing in compiler technology, program analysis, and formal methods. Experienced in designing semantic analysis pipelines, intermediate representations, and applying automated theorem proving to software verification. Additional experience in enterprise software development, workflow systems, and backend services. Seeking roles applying program analysis and formal reasoning to automated software development.

Experience

May 2025 - Present, *Founder & Research Software Engineer*, Karoo Compute Solutions

Designed and implemented Rose, an automated C-to-Rust translation system, in Rust.

- Rose web demo
- Semantic Lifting:
  - Designed and implemented semantic lifting for C and Rust programs
  - Defined multiple intermediate representations (IR) as part of the lifting pipeline
  - Implemented several existing and novel algorithms to transform code from one representation to the next
- Formal Reasoning:
  - Converted IR obtained from semantic lifting into SMT-LIB form
  - Used the Z3 theorem prover to determine whether two lifted programs are equivalent
- Built an LLM-assisted C-to-Rust translation prototype to verifiably translate a subset of both languages

Jan 2023 - Apr 2025, *Software Engineer*, EPI-USE Labs / EPI-USE Africa

Worked on FLOW, a Time and Attendance product implemented in ABAP:

- Built a framework to easily create OData (REST-based) services
- Built a workflow engine:
  - Designed to support arbitrary data, approvals, and triggering of actions (such as notifications)
  - Designed to enable client-specific workflow customization
  - Designed and implemented a Domain Specific Language (DSL) for compact and intuitive workflow configuration
  - Implemented a stack-based execution engine for boolean expressions

Transferred to EPI-USE Africa with the FLOW team in September 2024, following an internal group restructuring.

Education

2022, University of Pretoria

BSc Hons in Computer Science

2019 - 2021, University of Pretoria

BSc in Computer Science

Kaleb Bruwer
kaleb.bruwer@gmail.com
(+27)71 441 8189
Github
website

Software Development Skills

Languages

- **Expert:** Rust, C/C++, ABAP
- **Proficient:** Java, SQL
- **Familiar:** Python, JS

Technologies

- **Formal Methods:** Z3, SMT-LIB
- **Development:** Git, Linux

Achievements/Awards

2021, SANReN Cyber Security Challenge

- Capture the Flag Challenge between teams from various Southern African universities
- Our team placed second

2017, 2018, Gauteng North mathematics team

- Qualified for and competed in the ASSA South African Mathematics Team Competition
- In 2018, our team placed third nationally

Projects

2026, rprop: Propositional logic in Rust's type system

- A procedural-macro Rust library, published on crates.io
- Allow users to make logical propositions, claims, and write proofs that are checked at compile time

2020-2022, Spruce Goose: A C++ Minecraft server

- Github
- Implemented a Minecraft server in C++, compatible with the unmodified Minecraft client
- Networking with TCP sockets
- Multithreading and inter-thread communication
- Built data structures for efficiently handling 3D spatial data

2021, Unison: Smart Contract Verifier

- Final-year student project in team of five, formed part of Computer Science degree
- Created a DeFi escrow service
- Used blockchain/smart contract technology to track user agreements in a decentralized manner, where anonymous juries settled disputes
- Gained experience in developing software as a team, using Agile